F1 RACING ANALYTICS

Final Project Report

Project Name	F1 Racing Analytics
Team ID	
Project / Demo Link	See Appendix for published Tableau and GitHub links.

Prepared using the project details and Tableau views shared for this F1 analytics project.

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1. INTRODUCTION

1.1 Project Overview

F1 Racing Analytics is a data-visualisation project that explores Formula 1 performance using Tableau. It brings together driver points, team performance, championship trends, tyre-compound strategies, country-wise participation, podium finishes and race reliability into an interactive dashboard and a focused driver-points story.

1.2 Purpose

The purpose is to make F1 data easier to explore and compare. The dashboard presents high-level indicators and detailed views so a viewer can identify leading drivers and teams, compare performance across seasons, review podium outcomes and examine recorded DNF reasons.

2. IDEATION PHASE

2.1 Problem Statement

Formula 1 data can be spread across multiple fields and views, making it difficult to compare drivers, teams, seasons and race outcomes quickly. The project addresses this by presenting related performance indicators in one dashboard with a separate story view for driver points.

2.2 Empathy Map Canvas

Target users may include F1 fans, students and analysts who want to understand performance patterns without manually reviewing raw tables. They need clear rankings, meaningful labels, easy comparisons and filters that let them focus on a subset of drivers, teams or outcomes.

2.3 Brainstorming

The selected ideas were to create KPI cards, a driver points leaderboard, team performance ranking, team-by-season championship analysis, tyre-compound analysis, country-wise participation map, podium analysis and a driver reliability view for DNF reasons. A Top N control was used to make the leaderboard adjustable.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

A viewer opens the project home page, reads the overview, opens the dashboard to review the KPIs and visualisations, adjusts available filters, then opens the driver story to compare points using the Top N control. Published Tableau links and the GitHub repository provide access to the project outputs and code.

3.2 Solution Requirement

The solution requires the F1 dataset, Tableau worksheets and dashboard, a driver-points story, useful filters/parameters, readable visual titles, and a web page that links the dashboard and story. The web page should remain responsive and should clearly separate its home, dashboard and story sections.

3.3 Data Flow Diagram

F1 dataset → connect and inspect fields in Tableau → build calculations, filters and worksheets → assemble dashboard and story → publish to Tableau Public → embed or link published views from the project website → user explores the analysis.

3.4 Technology Stack

- Tableau Public for data exploration, visualisations, dashboard and story.
- HTML, CSS and JavaScript for the project website.
- Bootstrap for responsive layout and interface components.
- GitHub for source-code hosting and project versioning.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project responds to the need for a concise overview and deeper exploration of Formula 1 results. KPI cards summarise the dataset, ranking charts support comparisons, the season/team chart provides historical context, and the map and podium/tyre views broaden the analysis beyond points alone.

4.2 Proposed Solution

A Tableau dashboard containing nine main analytical areas, alongside a minimal driver-points story. The dashboard includes four KPI cards—Competing Teams (10), Seasons in Play (17), Race Outcomes Logged (6,464) and Circuits Showcased (29)—plus Driver Reliability – DNF Reasons, Driver Leaderboard Top 5 by Points, Team Performance Ranking, Team Championship Analysis, Tyre Compound Analysis, Country-wise Participation and Podium Analysis.

4.3 Solution Architecture

The dataset is used as the analytical source in Tableau. Worksheets provide the individual charts and KPI values; the dashboard combines these views and exposes relevant filters. The story presents the driver-points leaderboard and Top N control. Tableau Public hosts the published views, while the website acts as the presentation layer and links users to the analysis.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was completed in stages: (1) inspect the dataset and available fields, (2) create KPI and analytical worksheets, (3) build the dashboard, (4) configure the Top N driver leaderboard story, (5) publish the views to Tableau Public, and (6) create a website to present the project. Exact task dates and individual ownership were not provided, so they are not assumed in this report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was checked visually to confirm that its KPI cards and analytical views appeared in the dashboard layout. The driver story includes a Top N control. The published Tableau views may take time to load because they are externally hosted; actual load-time measurements and formal benchmark results were not provided.

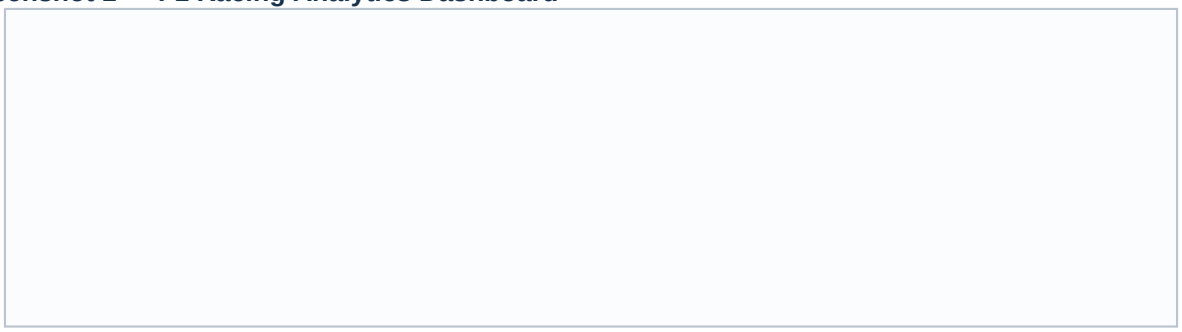
Test area	Project evidence / expected behaviour
Data rendered	KPI values and the dashboard's analytical views are present in the published dashboard.
Filters and parameter	Top N controls the number of drivers shown in the leaderboard; other Tableau filter cards are available on relevant views.
Visual layout	Dashboard sections are arranged vertically; chart sizing should be checked at the final browser width to avoid cramped labels.
Performance measurement	No measured load times, response-time benchmarks or browser/device test results were supplied.

7. RESULTS

7.1 Output Screenshots

Insert screenshots of the published dashboard, driver story, and website in the spaces below. This page has been intentionally reserved for screenshots.

Screenshot 1 — F1 Racing Analytics Dashboard



Screenshot 2 — Driver Points Leaderboard Story

Screenshot 3 — Project Website Home Page

8. ADVANTAGES & DISADVANTAGES

Advantages

• Combines several F1 performance perspectives in one place.
• KPI cards provide a quick overview of the data represented.
• Driver and team rankings make comparisons straightforward.
• The Top N control allows the driver leaderboard to be adjusted.
• Tableau Public links make the published views shareable.

Disadvantages / Limitations

• Embedded Tableau views depend on an internet connection and Tableau Public availability.
• Large dashboards may require scrolling and can feel cramped on smaller screens.
• No formal performance benchmark or comprehensive cross-device testing results were supplied.
• Interpretation is limited to the fields and records available in the dataset.

9. CONCLUSION

F1 Racing Analytics presents Formula 1 data through a set of Tableau KPI cards and visualisations, supported by a focused driver-points story. The project combines driver, team, season, tyre, geographic, podium and reliability perspectives in one analytical experience. The published Tableau views and GitHub repository make the project accessible for review.

10. FUTURE SCOPE

Potential improvements include adding season/team/driver cross-filter actions, improving mobile layout, adding concise chart explanations, measuring embed load time, validating the views across browsers, and refreshing the analysis as new race data becomes available. The website could also include a project methodology page and downloadable summary.

11. APPENDIX

Source Code

GitHub Repository: https://github.com/SOLORIS63/F1-Racing-Analysis.git

Dataset Link

Dataset source URL was not supplied in the current project details. Add the dataset URL here before final submission.

GitHub & Project Demo Links

GitHub: https://github.com/SOLORIS63/F1-Racing-Analysis.git

Published Tableau Dashboard: https://public.tableau.com/shared/9Z3NGD45Z

Published Tableau Driver Story: https://public.tableau.com/shared/JJ57Z8NDB

Project Demo Video Link

Paste the final video URL in the blank space below.

Video link

Team ID has been intentionally left blank. Replace the reserved screenshot boxes with your actual project screenshots before submission.